

1. VIDEO COLPOSCOPE

Item Code – OG001

Item Description – COLPOSCOPY SET

Technical Specifications

- 1) Latest HD Colposcope ·
- 2) Full HD Technology with CMOS Sensor ·
- 3) Minimum 1080p Resolution Sensor (1920x1080) ·
- 4) Aspect Ratio 16:9 ·
- 5) 42.8 Mega Pixels ·
- 6) No need of Dongle & Capture Card ·
- 7) Multiple Grades of green Filter ·
- 8) Recommended Lamp life 50000 hours ·
- 9) Highest quality 1-52x Magnification ·
- 10) Progressive Scanning Lines ·
- 11) USB output ·
- 12) Latest Digital Matrix Processor ·
- 13) Fine Advance & fast Auto/Manual focus-Fine Technology ·
- 14) E Flips & Mirror Image Facility ·
- 15) Display of genuine tissue colour and tone ·
- 16) Corner to Corner Uniform Brightness ·
- 17) Real time display of magnification ·
- 18) Acetic acid Test Timer ·
- 19) Electronic Green Filter with no light loss ·
- 20) Black Morphological Filter ·
- 21) Minimum illumination 0.1 Lux Multi-function Image processing software. ·
 - a) Integrated Management of View Examination, image capture & Freeze, Recording, Observation, Processing ·
 - b) Can be integrated to LAN and HIS ·
- 22) Reference pictures library to help user ·
- 23) Statistics mode with automatic flow chart facility is provided. ·
- 24) Printing of Multi Format photo report with Hammond graph Report ·
- 25) Powerful Comparison to compare ·
 - a) Colposcopy REID Evaluation ·
 - b) Report can be e-mailed as well
- 26) Accessories/Consumables:
 - a) Compatible Laptop Minimum 1KVA UPS

2. Cardiac Monitoring with Biphasic Defibrillator

Item Code – MED003

Item Description – Biphasic Defibrillator

Technical Specifications

- 1) The machine should have facility for ECG Monitoring, External Defibrillation, AED, & in-built recorder.
- 2) Should work on manual and automated external defibrillation (AED) in Bi-phasic mode. The maximum energy delivered by the device should be at least 200J or more.
- 3) Should monitor ECG through external paddles and monitoring electrodes.
- 4) Should have at least 6 inch or more color LCD display that provides clear visibility even in daylight.
- 5) Should have display of at least 4 waveforms of various monitoring parameters and their numerical data.
- 6) System should have instant boot up time, less than 5 seconds.
- 7) Should have external paddles with paddles contact indicator- for good paddle contact. Single adult and pediatric paddles should be available.
- 8) The unit should be capable of doing synchronized and asynchronized Cardio version.
- 9) Should have facility to continuously analyze patient ECG during CPR in AED mode.
- 10) Should have disposable pads with noiseless function to reduce the noise during CPR
- 11) The machine should have one key operation for easy switch over from adult to child mode in AED with same pads.
- 12) Should have facility of CPR device for actual CPR so that CPR can be performed in accordance with Guidelines. CPR device should provide audio visible guide of compression for adequate rate and depth.
- 13) Should have fast charging time, charging 200J within 5 seconds or less on both mains & battery.
- 14) Should have a battery capable of giving at least 70 discharges of maximum energy, 2 hours of continuous monitoring or 1 hour of pacing.
- 15) Should have both manual and automatic disarm facility.
- 16) The machine should be compact, portable with built in rechargeable battery, weight of the total machine should not be more than 7 Kg.
- 17) Should have status indicator that displays daily and monthly self-test results.
- 18) The machine must have capability for providing internal defibrillation shocks. The internal paddles of different sizes for adult, pediatric and infant patient should be available with the manufacturer.
- 19) Should display alarm message and have alarm indicator.
- 20) After defibrillation, the ECG waveform must recover within 5 seconds for immediately checking the result of defibrillation.
- 21) The machine should have Real time recording, Delayed recording, Event recording, Trend report up to 24 hour trend graph of HR, SpO₂, PR, CO₂, RR and VPC, Self-test result, Basic check result, Operation history/Device condition history.
- 22) The machine should have Automatic recording of Alarm report, Defibrillation recording, Periodic list report up to 24 hours of vital parameters.
- 23) The machine should have onscreen operation guide which shows the recommended measurement method for each parameter.
- 24) The machine should work on mains as well as on rechargeable battery.
- 25) The battery charging time should be less than 4 hours to full charge.
- 26) It should have selectable shock energy in AED mode.

- 27) The machine must be upgradeable for monitoring of vital sign parameter such as mainstream EtCO₂, SpO₂ and NIBP on installation site.

3. Operating Laproscope

Item Code – OG008

Item Description – 4K ICG Operating Laparoscopy Set with HD Camera and Monitor

Technical Specifications

- 1) 4K Resolution Medical Grade Monitor capable of displaying 3D.(32 inches) – QTY 1
 - a) 4K resolution 3D LCD Panel 32 inch (16:9 aspect ratio)
 - b) Should be capable of displaying 3840x2160 pixel resolution
 - c) Various signal inputs: Display port, 12G-SDI, HDMI and DVI
 - d) Viewing angle- Horizontal: 178 degrees, Vertical: 178 degrees
 - e) Contrast Ratio should be 1000:1
- 2) Camera Processor
- 3) Processor for following should be quoted.:
 - a) Processor for following should be quoted.: (2D 4K and 3D4K should be used at same processor)
 - b) 2 Dimensional endoscopic camera in 4K resolution (3840x2160)
 - c) 3 Dimensional endoscopic camera in 4K resolution (3840 x 2160)
 - d) 2 Dimensional endoscopic camera in Full HD resolution.
 - e) 3 Dimensional endoscopic camera for 3D exoscope.
 - f) System should have facility for Optical Contrast Differentiation System, and it Should have special filter for observation of capillary vessels and fine patterns in the superficial layer of mucosa for early detection of lesions.
 - g) System Should be capable of Near Infrared fluorescence Imaging (ICG application) with below features:
 - h) Overlay: White light image with superimposed display of NIR/ICG fluorescence.
- 4) Possible to select the preferred color for NIR/ICG imaging:
- 5) Either blue or green.
 - a) Monochromatic: NIR/ICG fluorescence signal in white. Background in black for maximum contrast.
 - b) Intensity Map: White light image with superimposed display of NIR/ICG fluorescence. NIR/ICG signal display will appear in different colors depending on the strength of the detected NIR signal.
 - c) Picture in Picture of visualization modes with Standard and Optical Contrast Differentiation.
 - d) The system should have facility of 3x Digital Zoom Lens or more.
 - e) The system should have facility of Inbuilt recording (video and still images) and can be controlled from camera head. If inbuilt recording facility not available, then recorder from the same manufacturer should be provided.
 - f) Automatic adjustment of light intensity of light source and controlled from Camera head.
 - g) Outputs: All Compatible outputs should be there (12GSDI, D.P)as a single output 4K resolution and DVI for HD resolution.

- h) System should be able to connect 3D Laparoscope (upgrade) in future.
- 6) 4K 2D Camera Head
- 7) Technical Specifications:
- 8) Pixels 3840 X 2160 Pixels
- 9) AGC: Microprocessor controlled
- 10) Lens: Integrated Zoom Lens $f = 19 \text{ mm}$
- 11) Color Space BT.2020 emulation
- 12) Control buttons: 3 (2 of them freely programmable).
 - a) Camera Head Should be able to perform both White light and Near Infrared application.
- 13) NIR LED Light source with Fiber optic cable
 - a) Should have Lumen >2000
 - b) Lamp life of approx. 30,000 hrs.
 - c) 4.8mm Fiber Optic Cable and 300cm Long.
 - d) Additional 4.8mm Fiber Optic Cable and 550cm Long.
 - e) with a sterilization tray for the fiber optic cable.
 - f) Should have touch display which provides an intuitive & user-friendly interface that directly displays relevant data
 - g) Lamp type: High-performance LEDs, white light LED and near infrared LED, which are active individually or simultaneously.
- 14) Certified To: - IEC 601-1 & UL 544 CE According to MDD, protection class 1/CF
- 15) TELESCOPES for 2D
- 16) Autoclavable Telescopes should be quoted as below for White light and Near Infrared application.
 - i) 5 mm 30 Degree with 29 cm or more working Length – 1 QTY.
- 17) 10mm 0 Degree with 30 cm or more working Length – 1 QTY
- 18) Equipment Cart from Same manufacturer Should be Quoted with.
- 19) CO2 Endoflator-50 ltr with pre & post heating with Gas heating & high flow mode with flow performance up to 50ltr/min. QTY-01
- 20) It should have following features:
 - a) Heatable insufflation tube for heating the CO2 gas up to patient body temperature.
 - b) Innovative Sensitive Mode for sensitive areas such as pediatric application with safety limits in the pressure up to maximum 15 mmhg and flow ranges up to maximum 15 l/min.
 - c) High flow mode with flow performance up to 50 l/min.
 - d) Individually programmable procedure list for starting device with personal presets.
 - e) Easy and intuitive use with user friendly colour touch screen for easy and precise setting of set values for pressure and flow and of insufflation mode, as well as for clear display of corresponding set values and actual values.
 - f) Optical and acoustic alarm signals in the event of patient overpressure.
 - g) Fully automatic, electronically controlled gas refill.
 - h) Safety system: Constant monitoring of intra-abdominal pressure; any overpressure is reduced immediately.
 - i) Control via Realistic User Interface.
- 21) Technical Specifications:
 - a) Gas flow: 0-50 l/min.
 - b) Pressure: 0-30 mmHg (4000Pa).

- c) High flow mode (0 to 50 l/min.)
 - d) Sensitive mode pressure 15 mmHg & flow 15 l/min for sensitive application
 - e) Electronic control and colour touch screen.
 - f) Following data are displayed on touch screen:
 - g) Insufflation mode.
 - h) Set value pressure (0-30 mmHg).
 - i) Current patient pressure.
 - j) Set value gas flow (0-50 l/min).
 - k) Current gas flow.
 - l) Gas consumption (0-999 l).
 - m) Status of gas heating.
 - n) Power supply: 100-240 VAC, 50/60 Hz.
 - o) Certified to: IEC 60601-1, CE label acc. to MDD, medical product Class IIb.
 - p) Type of protection against electrical shocks: Protection Class 1.
 - q) Degree of protection against electric shocks: Applied part type CF.
- 22) International protection code: IP21.
- 23) Irrigation Pump: Pressure regulated suction and irrigation system for use in Laparoscopy and Gynecology Qty : 1
- a) Unit should have touch screen as user interface.
 - b) Unit should have feature of automatically fixing maximum parameters for LAPAROSCOPY and HYSTEROSCOPY mode by the choice of tubing
 - c) Unit should have feature of simultaneous display of set values and actual values, enabling user for continuous monitoring of suction and irrigation parameters
 - d) Unit should have feature of use of disposable/reusable cassette tubing set for LAPAROSCOPY/HYSTEROSCOPY procedures.
 - e) Unit should be supplied with glass suction bottle 5 litres, bottle cap, bottle stand
 - f) Unit should have Pressure settings:
- 24) HYSTEROSCOPY mode: 0-200 mmHg
- 25) LAPAROSCOPY mode: 100/300/500 mmHg
- a) Unit should have Flow settings:
- 26) HYSTEROSCOPY mode: 200/400/600 ml/min
- 27) LAPAROSCOPY mode: 0-1300ml/min
- a) Suction Pressure regulated:
- 28) HYSTEROSCOPY mode: 0.1-(-)0.8bar (-80kPa)
- 29) LAPAROSCOPY mode: 0.1-(-)0.8bar(-80kPa)
- a) Power supply: 100-240VAC , 50/60HZ
 - b) Certified to: IEC 601-1, CE nach MDD
- 30) LAPAROSCOPIC GYNECOLOGY INSTRUMENTS SET
- 31) TROCAR, SIZE 11 MM, COLOR CODE: GREEN, CONSISTING OF: TROCAR ONLY, WITH PYRAMIDAL TIP CANNULA PYRAMIDAL TIP CANNULA WITHOUT VALVE, WITH INSUFFLATION STOP- COCK, LENGTH 10.5 CM MULTIFUNCTIONAL VALVE Qty: 2 WASHERS 2 PACKS
- 32) TROCAR, SIZE 6 MM, CONSISTING OF: TROCAR ONLY, WITH PYRAMIDAL TIP CANNULA WITHOUT VALVE, WITH 2 CANNULA WITHOUT VALVE, WITH

- INSUFFLATION STOP- COCK, LENGTH 10.5 CM N STOP- COCK, LENGTH 10.5 CM
MULTIFUNCTIONAL VALVE TIFUNCTIONAL VALVE Qty 6 WASHERS 5 PACKS
- 33) REDUCER, 11/5 MM: Qty 1
- 34) REDUCTION SLEEVE, 11/5 MM: Qty 1
- 35) KELLY DISSECTING AND GRASPING FORCEPS, ROTATING, WITH CONNECTOR
PIN FOR UNIPOLAR COAGULATION, SIZE 5 UNIPOLAR COAGULATION, SIZE 5
MM, LENGTH 36 CM, DOUBLE ACTION JAWS, CO 36 CM, DOUBLE ACTION JAWS,
CONSISTING OF: PLASTIC HANDLE, WITHOU PLASTIC HANDLE, WITHOUT
RATCHET, WITH LARGER CONTACT AREA ITH LARGER CONTACT AREA
OUTER TUBE, INSULATED FORCEPS Qty : 1
- 36) MANHES GRASPING FORCEPS, ROTAT ING, WITH CONNECTOR PIN FOR
UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 36 ULATION, SIZE 5 MM,
LENGTH 36 "CM,""TIGER-JAW"" , 2 X 4 TEETH, SINGLE " -JAW"" , 2 X 4 TEETH,
SINGLE" ACTION JAWS, CONSISTING OF: PLASTIC CONSISTING OF: PLASTIC
HANDLE, WITH MANHES STYLE RATCHET, WITH H MANHES STYLE RATCHET,
WITH LARGER CONTACT AREA OUTER TUBE, INS Qty: 1
- 37) DISSECTING AND GRASPING FORCEP S, ROTATING, WITH CONNECTOR PIN FOR
UNIPOLAR COAGULATION, SIZE 5 MM, L LAR COAGULATION, SIZE 5 MM,
LENGTH 36 CM, RIGHT ANGLED, DOUBLE ACTION RIGHT ANGLED, DOUBLE
ACTION JAWS, CONSISTING OF: PLASTIC HANDLE TING OF: PLASTIC HANDLE,
WITHOUT RATCHET, WITH LARGER CONTACT AR TCHET, WITH LARGER
CONTACT AREA OUTER TUBE, INSULATED Qty 1
- 38) METZENBAUM SCISSORS, ROTATING, DISMANTLING, WITH CONNECTOR PIN
FOR UNIPOLAR COAGULATION, WITH LUER-L OLAR COAGULATION, WITH
LUER-LOCK CONNECTOR FOR CLEANING, DOUBLE ACTION R FOR CLEANING,
DOUBLE ACTION JAWS, CURVED, LENGTH OF JAWS 15 MM, SIZE D, LENGTH OF
JAWS 15 MM, SIZE 5 MM, LENGTH 36 CM CONSISTING OF: 36 CM CONSISTING
OF: PLASTIC HANDLE, WITHOUT RATCHET, WITH LAR Qty 2
- 39) CROCE-OLMI GRASPING FORCEPS, R OTATING, WITH CONNECTOR PIN FOR
UNIPOLAR COAGULATION, SIZE 5 MM, LENGT COAGULATION, SIZE 5 MM,
LENGTH 36 CM, ATRAUMATIC, FENESTRATED, CURVED, AUMATIC,
FENESTRATED, CURVED, SINGLE ACTION JAWS, CONSISTING OF ON JAWS,
CONSISTING OF PLASTIC HANDLE, WITH MANHES RATCHET, WIT DLE, WITH
MANHES RATCHET, WITH LARGER CONTACT AREA OUTER TUBE Qty: 1
- 40) CLAW FORCEPS, ROTATING, SIZE 1 0 MM, LENGTH 36 CM, 2 X 3 TEETH, SINGLE
ACTION JAWS, CONSISTING OF: 33 CTION JAWS, CONSISTING OF METAL
HANDLE, WITH MANHES STYLE RATCH NDLE, WITH MANHES STYLE RATCHET
OUTER TUBE, INSULATED FS F ER TUBE, INSULATED FORCEPS INSERT T Qty1
- 41) GRASPING, FORCEPS, ROTATING, WITHOUT CONNECTOR PIN FOR UNIPOLAR
COAGULATION, SIZE 5 MM, LENGTH 36 CM, ATRAUMATIC, FENESTRATE ,
CURVED, SINGLE ACTION JAWS, CONSISTING OF: METAL HANDLE, WITHO UT
RATCHET, WITH LARGER CONTACT AREA OUTER TUBE, INSULATED FORCEPS
INSERT Qty 1
- 42) SUCTION AND IRRIGATION CANNULA WITH LATE RAL HOLES, SIZE 5 MM,
LENGTH 36 CM, FOR USE WITH SUCTION AND IRRIGATIO SE WITH SUCTION
AND IRRIGATION HANDLES: Qty1

- 43) SUCTION AND IRRIGATION CANNULA WITH LATERAL HOLES, SIZE 10 MM, LENGTH 36 CM, FOR USE WITH SUCTION AND IRRIGATION HANDLES. Qty 1
- 44) UNIPOLAR HIGH FREQUENCY CORD, WITH 4 MM PLUG, LENGTH 300 CM, FOR MODELS KARL STORZ, ERBE TYPE T, OLDER MODELS Z, ERBE TYPE T, OLDER MODELS AND ELLMAN Qty5
- 45) INJECTION NEEDLE, LUER-LOCK, DIAMETER 1.2 MM, SIZE 5 MM, LENGTH 36 CM. Qty1
- 46) MANHES HIGH FREQUENCY NEEDLE, FOR SPLITTING AND COAGULATION, INSULATED WITH CONNECTOR PIN FOR UNIPOLAR COAGULATOR PIN FOR UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 31 CM MM, LENGTH 31 CM Qty1
- 47) MYOMA FIXATION INSTRUMENT, SCREW SHAPED, Ø 5 MM Qty 2
- 48) CCL-EXTRACTOR, Ø 11 MM, WITH INSULATED BALL-SHAPED SPHERE, Ø 40 MM CONSISTING OF : 30103 A1 AUTOMATIC VALVE, SIZE 11 3010 3 V1 CANNULA Qty 1
- 49) ROBI® GRASPING FORCEPS, CLERMONT-FERRAND MODEL, ROTATING, DISMANTLING, WITH CONNECTOR PIN FOR BIPOLAR COAGULATOR PIN FOR BIPOLAR COAGULATION, WITH ESPECIALLY FINE ATRAUMATIC SPECIALLY FINE ATRAUMATIC SERRATION, FENESTRATED JAWS, DOUBLE ACTION JAW TRATED JAWS, DOUBLE ACTION JAWS, SIZE 5 MM, LENGTH 36 CM, COLOR CODE: M, LENGTH 36 CM, COLOR CODE: LIGHT BLUE CONSISTING OF: 38151 ROBI® PLA: Qty 1
- 50) KOH MACRO NEEDLE HOLDER, ERGONOMIC PISTOL HANDLE WITH DISENGAGEABLE RATCHET, RATCHET RELEASE ON THE LEFT SIDE, HET RELEASE ON THE LEFT SIDE, STRAIGHT JAWS, WITH TUNGSTEN CARBIDE INS WS, WITH TUNGSTEN CARBIDE INSERT Ø 5 MM, LENGTH 33 CM LENGTH 33 CM Qty 1
- 51) KOH MACRO NEEDLE HOLDER, ERGONOMIC AXIAL HANDLE WITH DISENGAGEABLE RATCHET, RATCHET RELEASE ON THE RIGHT SIDE, ET RELEASE ON THE RIGHT SIDE, LEFT CURVED JAWS, WITH TUNGSTEN CARBIDE JAWS, WITH TUNGSTEN CARBIDE INSERT Ø 5 MM, LENGTH 33 CM M, LENGTH 33 CM Qty1
- 52) UTERINE MANIPULATOR: UTERINE MANIPULATOR FOR PROVIDING TISSUE TENSION NEEDED FOR ANATOMICAL SURGERY. THE MANIPULATOR SHOULD HAVE APPLICATION IN TLH, LASH, LAVH AND HYSTERECTOMIES FOR MALIGNANT PATHOLOGIES: Qty 1
 - (1) THE MANIPULATOR SHOULD HAVE CAPACITY TO BE FIXED DIRECTLY TO THE PORTIO WITH INTEGRATED TENACULUM FORCEPS.
 - (2) THE MANIPULATOR SHOULD HAVE DIFFERENT CAP SIZES VARYING BETWEEN 28MM-43MM TO PREVENT GAS LOSS
- 53) 2.9 MM HYSTEROSCOPY SET
- 54) HYSTEROSCOPE TELESCOPE
- 55) Telescope 30°, ø 2.9 mm, length 30 cm, autoclavable, fiber optic light transmission incorporated, color code: black
- 56) DIAGNOSTIC HYSTEROSCOPE SHEATH

- 57) Hysteroscope sheath for diagnostic continuous irrigation size 4.5-4.6mm outer sheath and 3.8-3.9mm inner sheath for use with 2.9-3.0mm hysteroscope telescope.
- 58) OPERATIVE HYSTEROSCOPE SHEATH
- 59) Hysteroscope sheath for Operative continuous irrigation size 5.0-5.1mm outer sheath with operating channel 5Fr, leur lock adaptor and 4.3-4.4mm inner sheath for use with 2.9-3.0mm hysteroscope telescope. With stopcock & leur lock adaptor and inflow and outflow channel.. Oval shape to facilitate easy entry in internal cervical OS.
- 60) SEMI RIGID INSTRUMENTS FOR USE WITH OPERATING HYSTEROSCOPE: Qty 1 Each
- 61) SCISSOR
- 62) Scissors semi rigid, pointed jaws, 5 Fr., length 33-36cm, single action jaws, semi-rigid
- 63) BIOPSY & GRASPING FORCEP
- 64) Biopsy- and Grasping Forceps semi rigid, 5 Fr. , length 33-36cm, double action jaws
- 65) NEEDLE ELECTRODE
- 66) Bipolar Needle Electrode, 5 Fr., length 34 cm, unipolar
- 67) DISSECTION ELECTRODE BIPOLAR
- 68) Bipolar Dissection Electrode, 5Fr, 36cm, Bipolar
- 69) BIPOLAR CABLE
- 70) Bipolar cable
- 71) BIPOLAR CORD
- 72) Bipolar High Frequency, should fit in most of the standard electrosurgical unit having banana plug one side, length should be not less than 250cm.
 - (1) THE MANIPULATOR SHOULD HAVE VARYING DISTAL WORKING INSERTS IN VARYING 8 TO 10 LENGTH RANGING FROM 50MM TO 120MM. THE MANIPULATOR SHOULD HAVE FORCEP ATTACHMENT.
 - (2) THE MANIPULATOR SHOULD BE REUSABLE
- 73) CLEANING KIT FROM THE SAME MANUFACTURER: Qty 1
- 74) 22 FR. BIPOLAR RESECTOSCOPE
- 75) TELESCOPE
- 76) Telescope 12°, diameter 2.9 mm, length 30 cm, autoclavable, fiber optic light transmission incorporated,
- 77) BIPOLAR WORKING ELEMENT
- 78) Motion by means of a spring. The thumb support is movable. Return of the loop is controlled by the thumb and in rest position the electrode should rest inside the operating sheath, to be used with 2.9mm hysteroscopy telescope. Should work in saline
- 79) BIPOLAR CUTTING LOOP
- 80) BIPOLAR Cutting loop 21 Fr should work in saline
- 81) BIPOLAR CUTTING ELECTRODE POINTED
- 82) Cutting Electrode 21Fr, bipolar, pointed, should work in saline
- 83) BIPOLAR COAGULATING ELECTRODE BALL END
- 84) Coagulating Electrode 21Fr, bipolar, ball end should work in saline
- 85) RESECTOSCOPE SHEATH FOR BIIPOLAR
- 86) Continuous Flow Resectoscope Sheath 22 Fr., for Bi-Polar, including connection tubes for in- and outflow, 2 LUER-lock adaptors, 22Fr, oblique beak, rotating inner tube, with ceramic insulation, for use with working element. should work in saline
- 87) OBTURATOR

- 88) Obturator, for use with the Resectoscope sheath.
- 89) SINGLE PUNCTURE TUBAL LIGATION: Qty 1 Sets
- 90) Veress pneumoperitoneum needle spring- loaded blunt stylet LUER-lock, length 13 cm and 15 c.
- 91) Trocar, size 11 mm, pyramidal tip cannula out valve, insufflations stop-cock, length 10.5 cm, multifunctional valve: 04 Qty.
- 92) Laparoscope single puncture, Hopkins straight forward telescope 0 degree, parallel eyepiece, diameter 10 mm, length 27 cm, 6 mm instrument channel with adapter.
- 93) Ring applicator for use with operating single puncture laparoscopes Mounting cone & Pusher.

4. Hysteroscope operating

Item Code – OG009

**Item Description – 4K ICG Operating Hysteroscopy Set with HD Camera and Monitor
Laparoscopy Set with HD Vision & UHD Ready Platform**

Technical Specifications

- 1) 4K Resolution Medical Grade Monitor capable of displaying 3D.(32 inches) – QTY 1
 - a) 4K resolution 3D LCD Panel 32 inch (16:9 aspect ratio)
 - b) Should be capable of displaying 3840x2160 pixel resolution
 - c) Various signal inputs: Display port, 12G-SDI, HDMI and DVI
 - d) Viewing angle- Horizontal: 178 degrees, Vertical: 178 degrees
 - e) Contrast Ratio should be 1000:1
- 2) Camera Processor
- 3) Processor for following should be quoted.:
 - a) Processor for following should be quoted.: (2D 4K and 3D4K should be used at same processor)
 - b) 2 Dimensional endoscopic camera in 4K resolution (3840x2160)
 - c) 3 Dimensional endoscopic camera in 4K resolution (3840 x2160)
 - d) 2 Dimensional endoscopic camera in Full HD resolution.
 - e) 3 Dimensional endoscopic camera for 3D exoscope.
 - f) System should have facility for Optical Contrast Differentiation System, and it Should have special filter for observation of capillary vessels and fine patterns in the superficial layer of mucosa for early detection of lesions.
 - g) System Should be capable of Near Infrared fluorescence Imaging (ICG application) with below features:
 - h) Overlay: White light image with superimposed display of NIR/ICG fluorescence.
- 4) Possible to select the preferred color for NIR/ICG imaging:
- 5) Either blue or green.
 - a) Monochromatic: NIR/ICG fluorescence signal in white. Background in black for maximum contrast.
 - b) Intensity Map: White light image with superimposed display of NIR/ICG fluorescence. NIR/ICG signal display will appear in different colors depending on the strength of the detected NIR signal.

- c) Picture in Picture of visualization modes with Standard and Optical Contrast Differentiation.
- d) The system should have facility of 3x Digital Zoom Lens or more.
- e) The system should have facility of Inbuilt recording (video and still images) and can be controlled from camera head. If inbuilt recording facility not available, then recorder from the same manufacturer should be provided.
- f) Automatic adjustment of light intensity of light source and controlled from Camera head.
- g) Outputs: All Compatible outputs should be there (12GSDI, D.P) as a single output 4K resolution and DVI for HD resolution.
- h) System should be able to connect 3D Laparoscope (upgrade) in future.
- 6) 4K 2D Camera Head
- 7) Technical Specifications:
- 8) Pixels 3840 X 2160 Pixels
- 9) AGC: Microprocessor controlled
- 10) Lens: Integrated Zoom Lens f = 19 mm
- 11) Color Space BT.2020 emulation
- 12) Control buttons: 3 (2 of them freely programmable).
 - a) Camera Head Should be able to perform both White light and Near Infrared application.
- 13) NIR LED Light source with Fiber optic cable
 - a) Should have Lumen >2000
 - b) Lamp life of approx. 30,000 hrs.
 - c) 4.8mm Fiber Optic Cable and 300cm Long.
 - d) Additional 4.8mm Fiber Optic Cable and 550cm Long.
 - e) with a sterilization tray for the fiber optic cable.
 - f) Should have touch display which provides an intuitive & user-friendly interface that directly displays relevant data
 - g) Lamp type: High-performance LEDs, white light LED and near infrared LED, which are active individually or simultaneously.
- 14) Certified To: - IEC 601-1 & UL 544 CE According to MDD, protection class 1/CF
- 15) TELESCOPES for 2D
- 16) Autoclavable Telescopes should be quoted as below for White light and Near Infrared application.
 - i) 5 mm 30 Degree with 29 cm or more working Length – 1 QTY.
- 17) 10mm 0 Degree with 30 cm or more working Length – 1 QTY
- 18) Equipment Cart from Same manufacturer Should be Quoted with.
- 19) CO2 Endoflator-50 ltr with pre & post heating with Gas heating & high flow mode with flow performance up to 50ltr/min. QTY-01
- 20) It should have following features:
 - a) Heatable insufflation tube for heating the CO2 gas up to patient body temperature.
 - b) Innovative Sensitive Mode for sensitive areas such as pediatric application with safety limits in the pressure up to maximum 15 mmhg and flow ranges up to maximum 15 l/min.
 - c) High flow mode with flow performance up to 50 l/min.
 - d) Individually programmable procedure list for starting device with personal presets.

- e) Easy and intuitive use with user friendly colour touch screen for easy and precise setting of set values for pressure and flow and of insufflation mode, as well as for clear display of corresponding set values and actual values.
- f) Optical and acoustic alarm signals in the event of patient overpressure.
- g) Fully automatic, electronically controlled gas refill.
- h) Safety system: Constant monitoring of intra-abdominal pressure; any overpressure is reduced immediately.
- i) Control via Realistic User Interface.

21) Technical Specifications:

- a) Gas flow: 0-50 l/min.
- b) Pressure: 0-30 mmHg (4000Pa).
- c) High flow mode (0 to 50 l/min.)
- d) Sensitive mode pressure 15 mmhg& flow 15 l/min for sensitive application
- e) Electronic control and colour touch screen.
- f) Following data are displayed on touch screen:
- g) Insufflation mode.
- h) Set value pressure (0-30 mmHg).
- i) Current patient pressure.
- j) Set value gas flow (0-50 l/min).
- k) Current gas flow.
- l) Gas consumption (0-999 l).
- m) Status of gas heating.
- n) Power supply: 100-240 VAC, 50/60 Hz.
- o) Certified to: IEC 60601-1, CE label acc. to MDD, medical product Class IIb.
- p) Type of protection against electrical shocks: Protection Class 1.
- q) Degree of protection against electric shocks: Applied part type CF.

22) International protection code: IP21.

23) Irrigation Pump: Pressure regulated suction and irrigation system for use in Laparoscopy and Gynecology Qty : 1

- a) Unit should have touch screen as user interface.
- b) Unit should have feature of automatically fixing maximum parameters for LAPAROSCOPY and HYSTEROSCOPY mode by the choice of tubing
- c) Unit should have feature of simultaneous display of set values and actual values, enabling user for continuous monitoring of suction and irrigation parameters
- d) Unit should have feature of use of disposable/reusable cassette tubing set for LAPAROSCOPY/HYSTEROSCOPY procedures.
- e) Unit should be supplied with glass suction bottle 5 litres, bottle cap, bottle stand
- f) Unit should have Pressure settings:

24) HYSTEROSCOPY mode: 0-200 mmHg

25) LAPAROSCOPY mode: 100/300/500 mmHg

- a) Unit should have Flow settings:

26) HYSTEROSCOPY mode: 200/400/600 ml/min

27) LAPAROSCOPY mode: 0-1300ml/min

- a) Suction Pressure regulated:

28) HYSTEROSCOPY mode: 0.1-(-)0.8bar (-80kPa)

29) LAPAROSCOPY mode: 0.1-(-)0.8bar(-80kPa)

- a) Power supply: 100-240VAC , 50/60HZ
- b) Certified to: IEC 601-1, CE nach MDD
- 30) LAPAROSCOPIC GYNECOLOGY INSTRUMENTS SET
- 31) TROCAR, SIZE 11 MM, COLOR CODE: GREEN, CONSISTING OF: TROCAR ONLY, WITH PYRAMIDAL TIP CANNULA PYRAMIDAL TIP CANNULA WITHOUT VALVE, WITH INSUFFLATION STOP- COCK, LENGTH 10.5 CM MULTIFUNCTIONAL VALVE Qty: 2 WASHERS 2 PACKS
- 32) TROCAR, SIZE 6 MM, CONSISTING OF: TROCAR ONLY, WITH PYRAMIDAL TIP CANNULA WITHOUT VALVE, WITH 2 CANNULA WITHOUT VALVE, WITH INSUFFLATION STOP- COCK, LENGTH 10.5 CM INSUFFLATION STOP- COCK, LENGTH 10.5 CM MULTIFUNCTIONAL VALVE MULTIFUNCTIONAL VALVE Qty 6 WASHERS 5 PACKS
- 33) REDUCER, 11/5 MM: Qty 1
- 34) REDUCTION SLEEVE, 11/5 MM: Qty 1
- 35) KELLY DISSECTING AND GRASPING FORCEPS, ROTATING, WITH CONNECTOR PIN FOR UNIPOLAR COAGULATION, SIZE 5 MM, UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 36 CM, DOUBLE ACTION JAWS, CO 36 CM, DOUBLE ACTION JAWS, CONSISTING OF: PLASTIC HANDLE, WITHOUT PLASTIC HANDLE, WITHOUT RATCHET, WITH LARGER CONTACT AREA WITH LARGER CONTACT AREA OUTER TUBE, INSULATED FORCEPS Qty : 1
- 36) MANHES GRASPING FORCEPS, ROTATING, WITH CONNECTOR PIN FOR UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 36 CM, UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 36 CM, "TIGER-JAW", 2 X 4 TEETH, SINGLE " -JAW", 2 X 4 TEETH, SINGLE " ACTION JAWS, CONSISTING OF: PLASTIC HANDLE, WITH MANHES STYLE RATCHET, WITH H MANHES STYLE RATCHET, WITH LARGER CONTACT AREA OUTER TUBE, INS Qty: 1
- 37) DISSECTING AND GRASPING FORCEPS, ROTATING, WITH CONNECTOR PIN FOR UNIPOLAR COAGULATION, SIZE 5 MM, UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 36 CM, RIGHT ANGLED, DOUBLE ACTION RIGHT ANGLED, DOUBLE ACTION JAWS, CONSISTING OF: PLASTIC HANDLE CONSISTING OF: PLASTIC HANDLE, WITHOUT RATCHET, WITH LARGER CONTACT AREA WITH RATCHET, WITH LARGER CONTACT AREA OUTER TUBE, INSULATED Qty 1
- 38) METZENBAUM SCISSORS, ROTATING, DISMANTLING, WITH CONNECTOR PIN FOR UNIPOLAR COAGULATION, WITH LUER-L LOCK FOR UNIPOLAR COAGULATION, WITH LUER-LOCK CONNECTOR FOR CLEANING, DOUBLE ACTION RATCHET FOR CLEANING, DOUBLE ACTION JAWS, CURVED, LENGTH OF JAWS 15 MM, SIZE D, LENGTH OF JAWS 15 MM, SIZE 5 MM, LENGTH 36 CM CONSISTING OF: 36 CM CONSISTING OF: PLASTIC HANDLE, WITHOUT RATCHET, WITH LARGER Qty 2
- 39) CROCE-OLMI GRASPING FORCEPS, ROTATING, WITH CONNECTOR PIN FOR UNIPOLAR COAGULATION, SIZE 5 MM, UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 36 CM, ATRAUMATIC, FENESTRATED, CURVED, ATRAUMATIC, FENESTRATED, CURVED, SINGLE ACTION JAWS, CONSISTING OF: PLASTIC HANDLE, WITH MANHES RATCHET, WITH MANHES RATCHET, WITH LARGER CONTACT AREA OUTER TUBE Qty: 1
- 40) CLAW FORCEPS, ROTATING, SIZE 10 MM, LENGTH 36 CM, 2 X 3 TEETH, SINGLE ACTION JAWS, CONSISTING OF: 36 CM ACTION JAWS, CONSISTING OF METAL

- HANDLE, WITH MANHES STYLE RATCH NDLE, WITH MANHES STYLE RATCHET
OUTER TUBE, INSULATED FS F ER TUBE, INSULATED FORCEPS INSERT T Qty1
- 41) GRASPING, FORCEPS, ROTATING, WITHOUT CONNECTOR PIN FOR UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 36 CM, ATRAUMATIC, FENESTRATE , CURVED, SINGLE ACTION JAWS, CONSISTING OF: METAL HANDLE, WITHO UT RATCHET, WITH LARGER CONTACT AREA OUTER TUBE, INSULATED FORCEPS INSERT Qty 1
 - 42) SUCTION AND IRRIGATION CANNULA WITH LATE RAL HOLES, SIZE 5 MM, LENGTH 36 CM, FOR USE WITH SUCTION AND IRRIGATION SE WITH SUCTION AND IRRIGATION HANDLES: Qty1
 - 43) SUCTION AND IRRIGATION CANNULA WITH LATERAL HOLES, SIZE 10 MM, LENGTH 36 CM, FOR USE WITH SUCTION AND IRRIGATION HANDLES. Qty 1
 - 44) UNIPOLAR HIGH FREQUENCY CORD, WITH 4 MM PLUG, LENGTH 300 CM, FOR MODELS KARL STORZ, ERBE TYPE T, OLDER MODELS Z, ERBE TYPE T, OLDER MODELS AND ELLMAN Qty5
 - 45) INJECTION NEEDLE, LUER-LOCK, DIAMETER 1. 2 MM, SIZE 5 MM, LENGTH 36 CM. Qty1
 - 46) MANHES HIGH FREQUENCY NEEDLE, FOR SPLITT ING AND COAGULATION, INSULATED WITH CONNECTOR PIN FOR UNIPOLAR COAGULA CTOR PIN FOR UNIPOLAR COAGULATION, SIZE 5 MM, LENGTH 31 CM MM, LENGTH 31 CM Qty1
 - 47) MYOMA FIXATION INSTRUMENT, SCREW SHAPED, Ø 5 MM Qty 2
 - 48) CCL-EXTRACTOR, Ø 11 MM, WITH INSULATED BALL-SHAPED SPHERE, Ø 40 MM CONSISTING OF : 30103 A1 AUTOMATIC VALVE, SIZE 11 3010 3 V1 CANNULA Qty 1
 - 49) ROBI® GRASPING FORCEPS, CLERMONT-FERRAND MODEL, ROTATING, DISMANTLING, WITH CONNECTOR PIN FOR BIPOLAR COAGULAT CTOR PIN FOR BIPOLAR COAGULATION, WITH ESPECIALLY FINE ATRAUMATIC SERR PECIALLY FINE ATRAUMATIC SERRATION, FENESTRATED JAWS, DOUBLE ACTION JA TRATED JAWS, DOUBLE ACTION JAWS, SIZE 5 MM, LENGTH 36 CM, COLOR CODE: M, LENGTH 36 CM, COLOR CODE: LIGHT BLUE CONSISTING OF: 38151 ROBI® PLA: Qty 1
 - 50) KOH MACRO NEEDLE HOLDER, ERGONOMIC PISTO L HANDLE WITH DISENGAGEABLE RATCHET, RATCHET RELEASE ON THE LEFT SIDE, HET RELEASE ON THE LEFT SIDE, STRAIGHT JAWS, WITH TUNGSTEN CARBIDE INS WS, WITH TUNGSTEN CARBIDE INSERT Ø 5 MM, LENGTH 33 CM LENGTH 33 CM Qty 1
 - 51) KOH MACRO NEEDLE HOLDER, ERGONOMIC AXIAL HANDLE WITH DISENGAGEABLE RATCHET, RATCHET RELEASE ON THE RIGHT SIDE, ET RELEASE ON THE RIGHT SIDE, LEFT CURVED JAWS, WITH TUNGSTEN CARBIDE JAWS, WITH TUNGSTEN CARBIDE INSERT Ø 5 MM, LENGTH 33 CM M, LENGTH 33 CM Qty1
 - 52) UTERINE MANIPULATOR: UTERINE MANIPULATOR FOR PROVIDING TISSUE TENSION NEEDED FOR ANATOMICAL SURGERY. THE MANIPULATOR SHOULD HAVE APPLICATION IN TLH, LASH, LAVH AND HYSTERECTOMIES FOR MALIGNANT PATHOLOGIES: Qty 1

- (1) THE MANIPULATOR SHOULD HAVE CAPACITY TO BE FIXED DIRECTLY TO THE PORTIO WITH INTEGRATED TENACULUM FORCEPS.
- (2) THE MANIPULATOR SHOULD HAVE DIFFERENT CAP SIZES VARYING BETWEEN 28MM-43MM TO PREVENT GAS LOSS
- 53) 2.9 MM HYSTEROSCOPY SET
- 54) HYSTEROSCOPE TELESCOPE
- 55) Telescope 30°, ø 2.9 mm, length 30 cm, autoclavable, fiber optic light transmission incorporated, color code: black
- 56) DIAGNOSTIC HYSTEROSCOPE SHEATH
- 57) Hysteroscope sheath for diagnostic continuous irrigation size 4.5-4.6mm outer sheath and 3.8-3.9mm inner sheath for use with 2.9-3.0mm hysteroscope telescope.
- 58) OPERATIVE HYSTEROSCOPE SHEATH
- 59) Hysteroscope sheath for Operative continuous irrigation size 5.0-5.1mm outer sheath with operating channel 5Fr, leur lock adaptor and 4.3-4.4mm inner sheath for use with 2.9-3.0mm hysteroscope telescope. With stopcock & leur lock adaptor and inflow and outflow channel.. Oval shape to facilitate easy entry in internal cervical OS.
- 60) SEMI RIGID INSTRUMENTS FOR USE WITH OPERATING HYSTEROSCOPE: Qty 1 Each
- 61) SCISSOR
- 62) Scissors semi rigid, pointed jaws, 5 Fr., length 33-36cm, single action jaws, semi-rigid
- 63) BIOPSY & GRASPING FORCEP
- 64) Biopsy- and Grasping Forceps semi rigid, 5 Fr. , length 33-36cm, double action jaws
- 65) NEEDLE ELECTRODE
- 66) Bipolar Needle Electrode, 5 Fr., length 34 cm, unipolar
- 67) DISSECTION ELECTRODE BIPOLAR
- 68) Bipolar Dissection Electrode, 5Fr, 36cm, Bipolar
- 69) BIPOLAR CABLE
- 70) Bipolar cable
- 71) BIPOLAR CORD
- 72) Bipolar High Frequency, should fit in most of the standard electrosurgical unit having banana plug one side, length should be not less than 250cm.
 - (1) THE MANIPULATOR SHOULD HAVE VARYING DISTAL WORKING INSERTS IN VARYING 8 TO 10 LENGTHRANGING FROM 50MM TO 120MM. THE MANIPULATOR SHOULD HAVE FORCEP ATTACHEMENT.
 - (2) THE MANIPULATOR SHOULD BE REUSABLE
- 73) CLEANING KIT FROM THE SAME MANUFACTURER: Qty 1
- 74) 22 FR. BIPOLAR RESECTOSCOPE
- 75) TELESCOPE
- 76) Telescope 12°, diameter 2.9 mm, length 30 cm, autoclavable, fiber optic light transmission incorporated,
- 77) BIPOLAR WORKING ELEMENT
- 78) Motion by means of a spring. The thumb support is movable. Return of the loop is controlled by the thumb and in rest position the electrode should rest inside the operating sheath, to be used with 2.9mm hysteroscopy telescope. Should work in saline
- 79) BIPOLAR CUTTING LOOP

- 80) BIPOLAR Cutting loop 21 Fr should work in saline
- 81) BIPOLAR CUTTING ELECTRODE POINTED
- 82) Cutting Electrode 21Fr, bipolar, pointed, should work in saline
- 83) BIPOLAR COAGULATING ELECTRODE BALL END
- 84) Coagulating Electrode 21Fr, bipolar, ball end should work in saline
- 85) RESECTOSCOPE SHEATH FOR BIIPOLAR
- 86) Continuous Flow Resectoscope Sheath 22 Fr., for Bi-Polar, including connection tubes for in- and outflow, 2 LUER-lock adaptors, 22Fr, oblique beak, rotating inner tube, with ceramic insulation, for use with working element. should work in saline
- 87) OBTURATOR
- 88) Obturator, for use with the Resectoscope sheath.
- 89) SINGLE PUNCTURE TUBAL LIGATION: Qty 1 Sets
- 90) Veress pneumoperitoneum needle spring- loaded blunt stylet LUER-lock, length 13 cm and 15 c.
- 91) Trocar, size 11 mm, pyramidal tip cannula out valve, insufflations stop-cock, length 10.5 cm, multifunctional valve: 04 Qty.
- 92) Laparoscope single puncture, Hopkins straight forward telescope 0 degree, parallel eyepiece, diameter 10 mm, length 27 cm, 6 mm instrument channel with adapter.
- 93) Ring applicator for use with operating single puncture laparoscopes Mounting cone & Pusher.

5. Hysteroscope office

Item Code – OG010

Item Description – OFFICE HYSTEROSCOPES

Technical Specifications

- 1) High definition Camera System
- 2) The system should have the maximum Resolution of 1920 X 1080 pixels, progressive scan and the consistent use of 16: 9 formats
- 3) Internal Medical grade recorder for Capturing FULL HD Videos and HD Stills with storage capability of 50GB internally and 500 GB externally.
- 4) 18 inch or more touch screen for Play back of Recorded Videos and Controlling the Features camera system.
- 5) System should have facility of controlling the light source and recording device from the camera head.
- 6) Video Scopes can be connected to the same.
- 7) Technical Specifications:
- 8) Image sensor: CMOS
- 9) Pixels 1920 x 1080
- 10) AGC: Microprocessor controller
- 11) Control buttons: 3 (2 of them freely programmable)
- 12) Input: Keyboard input for character generator.
- 13) Power Supply:- 100-240 VAC 50/60 Hz

- 14) Certified to : IEC 601-1, 601-2-18, CSA 22.2 No. 601, UL 2601 and CE according to MDD, protection class I/CF
- 15) High Definition Medical Grade Monitor:
- 16) The monitor should have:
- 17) HDTV display in original 16: 9 format.
- 18) 1080 p/ 50 & 1080 p/60 displays possible.
- 19) LED crystal display.
- 20) Max. Resolution of 1920X1080.
- 21) Screen diagonal – 18 inches or more
- 22) Monitor must be inbuilt
- 23) LED Light Source with Fiber optic cable
- 24) Color Temperatures 6400K
- 25) Inbuilt
- 26) Light Outlets – 1
- 27) Light Intensity Adjustment :- Continuously adjustable manually and automatically by cameras video output signal.
- 28) Should be supplied with Diameter 4.8mm, Length 300cm.
- 29) Certified To :- IEC 601-1 & UL 544 CE According to MDD , protection class I/CF
- 30) Note: Light source should be inbuilt with processor and Monitor.
- 31) Suction Irrigation Pump
- 32) Pressure regulated suction and irrigation system for use in laparoscopy and hysteroscopy.
Unit should be with touch screen as user interface. The unit should be with feature of automatically fixing maximum parameters for LAPROSCOPY and HYSTEROSCOPY mode by the choice of tubing
- 33) Unit should have feature of simultaneous display of set values and actual values, enabling user for continuous monitoring of suction and irrigation parameters.
- 34) Should have feature of use of disposable/reusable cassette tubing set for LAPROSCOPY/HYSTEROSCOPY procedures.
- 35) Unit should be supplied with glass suction bottle 5lt, bottle cap, bottle stand and bottle holder.
- 36) Pressure Flow:
- 37) HYS 0-200 mmHg
- 38) LAP 100/300/500 Hg
- 39) Flow:
- 40) LAP: 0-1300ml/min
- 41) HYS: 200/400/600 ml/min
- 42) Suction Pressure regulated:
- 43) HYST: 0.1-(-)0.8bar (-80kPa)
- 44) LAP: 0.1-(-)0.8bar(-80kPa)
- 45) Power supply: 100-240VAC , 50/60HZ
- 46) Certified to: IEC 601-1, CE nach MDD
- 47) Hysteroscope - Diagnostic & Operative
- 48) Telescope 30°, ø 2.9 mm, length 30 cm with 30X magnification, autoclavable, fiber optic light transmission incorporated, color code: black

- 49) Inner sheath, size 4.3 - 4.4 mm with channel for semirigid 5 Fr. Instruments, with 1 stopcock and 1 LUER-Lock adapter, for use with outer sheath and 2.9mm - 3.0 mm Hysteroscope telescope
- 50) Outer sheath, size 5.0 - 5.1 mm with 1 stopcock and 1 LUER-Lock adapter, for use with Inner sheath, Oval shape to facilitate easy entry in internal cervical OS.
- 51) Scissors semi rigid, blunt tips, 5 Fr., length 36 - 40cm, single action jaws: 2 qty
- 52) Biopsy- and Grasping Forceps semi rigid, 5 Fr. , length 36 - 40cm, double action jaws: 2 qty
- 53) Bipolar Vaporization Electrode, Semirigid, 5 Fr., Needle Electrode Angled 90°, Length 42cm: 2 qty
- 54) Bipolar High Frequency, should fit in most of the standard electrosurgical unit having banana plug one side, length should be not less than 250cm.: 2 qty
- 55) Should have a separate sterilization/autoclavable tray for telescope.
- 56) Equipment Cart from the same manufacturer.

6. ABG Machine

Item Code – CCM013

Item Description – ARTERIAL BLOOD GAS (ABG) ANALYZER

Technical Specifications

- 1) Essential Measured parameters; pH, pCO₂, pO₂, tHb, Barometric Pressure, Lactate, Na⁺, K⁺, Ca⁺⁺, Cl⁻. All these parameters should be measured simultaneously.
- 2) Should have minimum 15 calculated parameters including SaO₂, Bicarbonate (HCO₃), Standard HCO₃, Base Excess of Blood (BE), Base Excess of extra cellular fluid.
- 3) Sample volume less than 100 ul.
- 4) Should have minimum process time (less than 5 min).
- 5) Warm up time should be less than 30 minutes.
- 6) Maintenance free electrodes.
- 7) Fully automatic liquid calibration of all parameters at user-defined intervals.
- 8) Should be with numeric keypad, graphic/LCD display, and inbuilt printer.
- 9) Should have interface for PC compatibility.
- 10) QC should be based on test parameters.
- 11) Automatic result processing, test ordering and provision for bi-directional LIS interface should be available.
- 12) Automatic data archiving and customizable layout.
- 13) Should have provision for data backup.
- 14) LCD/Graphical Display in-built.
- 15) PHYSICAL CHARACTERISTICS Noise pressure level: ≤ 60 dB.
- 16) Should maintain nominal temperature and the heat should be disbursed through a cooling mechanism.
- 17) Portable.
- 18) ENERGY SOURCE 220+/-10% VAC, 50 Hz.
- 19) At least 30 minutes backup.
- 20) ENVIRONMENTAL AND DEPARTMENTAL CONSIDERATIONS Capable of operating continuously in ambient temperature of -10 to 60 deg C and relative humidity of up to 90% in ideal circumstances.
- 21) Accessories/Consumables:
 - a) Reagents for minimum 200 tests should be provided along with the machine.

- b) Electrodes for all the parameters specified – 01 set.
- c) Quality control tools/reagents for minimum 200 tests or as per requirement.

7. Vessel sealer

Item Code – CTVS013

Item Description – Advance Electrosurgical Unit with Vessel Sealing and Bipolar Dissection

1. Closed-loop, multi-parameter tissue-sensing feedback for consistent output across variable tissue impedances.
2. Touchscreen GUI $\geq 5"$ color with clear readouts of active mode, set power, real-time output/impedance, pad status, and alarms.
3. Programmable memory ≥ 100 user presets with one-touch recall; password-protected service menu; event/error logs.
4. Activation: hand-switch pencils and multi-pedal footswitch (independent pedals for Cut/Coag/Bipolar/Seal; water-resistant, medical-grade).
5. Outputs: minimum two monopolar output ports and one bipolar port; dedicated vessel sealing activation (hand or foot); simultaneous connection with clear channel selection.
6. Language: UI in English (Hindi/other local language preferred).
- 1) **Monopolar CUT (HF ~ 390 kHz or equivalent)**
 - a) Modes (≥ 7): Pure Cut, Blend 1, Blend 2, Blend 3, Endo-Cut 1, Endo-Cut 2 (or equivalent pulsed cut), Laparoscopic Fine/Precise Cut.
 - b) Smooth incision in dry or saline-splashed field; adjustable starting burst for endoscopic cutting.
- 3) **Monopolar COAGULATION**
 - a) 9. Modes (≥ 3): Soft/Desiccate, Forced/Standard, Spray/Fulgurate (randomized spark suppression to limit carbonization).
 - b) 10. Adjustable effect levels; stable coag in presence of irrigation/saline mist.
- 4) **BIPOLAR**
 - a) 11. Modes (≥ 3): Micro (fine), Macro (standard), Precise/Soft (for neurosurgery/ENT).
 - b) 12. Saline plasma bipolar capability for resection, cutting and coagulation in isotonic saline (TURP/TCRE/arthroscopy) with dedicated resection profiles.
- 5) **ADVANCED VESSEL SEALING (Integrated)**
 - a) 13. Seal capability: Blood vessels and tissue bundles up to 7 mm outer diameter.
 - b) 14. Energy delivery: Closed-loop, pulsed RF with Fine-Seal and Rapid-Seal (or equivalent) profiles; ≥ 5 effect levels.
 - c) 15. Max sealing power: ≥ 150 W @ 100Ω (or equivalent performance) with target crest factor ≈ 1.5 in sealing mode.
 - d) 16. End-point control: Auto-stop at seal completion with audible/visual confirmation; alarm on improper/incomplete seal.
 - e) 17. Safety prompts: Tissue re-grasp/re-hold prompts; over-temperature, over-current and over-time protections.
 - f) 18. Compatible with reusable and single-use open and laparoscopic sealing instruments (5 mm & 10 mm), with or without integrated mechanical divider/knife.
- 6) **Power & Performance**

- a) 19. Total generator rating: ≥ 400 W continuous (at nominal mains), thermally protected for surgical duty cycles.
- b) 20. Output stability: Closed-loop regulation across 5–135 Ω tissue impedance (or wider), with automatic arc control.
- c) 21. Return Electrode Monitoring (REM/CQM): Continuous monitoring of split pad contact; audible/visual alarm for poor contact/lead off; output inhibit until safe pad contact restored.
- d) 22. Self-test: Automatic power-on diagnostics for output stages, sensors, footswitch, UI, and alarms.

7) Electrical & Environmental

- a) 23. Mains: 180–265 VAC, 50/60 Hz; max input ≤ 1000 VA; IEC-320 inlet with hospital-grade power cord.
- b) 24. UPS compatibility: Operable with 1.5 kVA line-interactive/online UPS.
- c) 25. Operating conditions: 10–40 °C, 15–90% RH non-condensing; altitude up to 3000 m.

8) Safety Features (Mandatory)

- a) 26. Output inhibit on fault; HF leakage per IEC 60601-2-2.
- b) 27. Audible tones differentiated for Cut/Coag/Seal; volume adjustable; visual indicators near active connectors.
- c) 28. Patient return electrode/adaptor compatibility with adult & pediatric split pads.
- d) 29. Floating outputs (CF-type) for bipolar modes; patient circuit isolation meeting IEC requirements.

9) Accessories & Deliverables (Minimum)

- a) 30. Main ESU with vessel sealing generator – 1 no.
- b) 31. Footswitch, multi-pedal (min. 3 pedals; sealed, non-slip) – 1 no.
- c) 32. Monopolar hand-switch pencils with 3-m cables – 6 nos (sterile, disposable tips: blades/needle/ball – assorted ≥ 50 pcs).
- d) 33. Patient return electrodes (split-type) – ≥ 50 adult, ≥ 10 pediatric with compatible cables.
- e) 34. Bipolar forceps, reusable – 2 nos (one straight, one bayonet), with bipolar cable – 1 no.
- f) 35. Vessel sealing instruments (compatibility with 5 mm & 10 mm): Laparoscopic 5 mm, length 37–45 cm, curved/Maryland jaw, reusable – 2 nos.
- g) 36. Open sealer, 20–24 cm, curved jaw, reusable – 1 no.
- h) 37. Compatible reusable sealing cable – 1 no.
- i) 38. Starter set of single-use sealers (if applicable) – 10 nos (5 mm laparoscopic).
- j) 39. Saline plasma bipolar resection kit compatibility (loops/electrodes/working elements) to be demonstrated; include initial consumables pack suitable for TURP/TCRE – 1 set.
- k) 40. Cart/trolley with locking castors, isolation transformer shelf and cable management – 1 no (preferred).
- l) 41. IFU/user manual, service manual, circuit diagrams, preventive maintenance checklist (English; soft + hard copy).

8. Boyle's Anesthesia Workstation

Item Code – CARDIO 020

Item Description – ANAESTHESIA WORKSTATION

Technical Specifications

- 1) Gas Delivery System
 - a) Provision for delivery of: Oxygen (O₂), Nitrous Oxide (N₂O), and Medical Air
 - b) Monitoring: Pressure gauges/ventilator screen for:
- 2) O₂ and N₂O from cylinder
- 3) O₂, N₂O, and AIR from central pipeline
 - a) Gas Supply Attachments:
- 4) Independent connections for central pipeline supply and pin-indexed cylinders
- 5) PIN Indexed Yoke System for O₂ & N₂O
- 6) DISS/Non-Interchangeable Screw Thread (NIST) system for pipeline inlets (O₂, N₂O, Air)
 - a) Hypoxic Guard System:
- 7) Oxygen and Nitrous Oxide are linked to ensure minimum 21–25% oxygen delivery
- 8) N₂O Cut-off if oxygen supply fails
- 9) Flowmeter & Oxygen Flush
 - a) Three-Gas System: O₂, N₂O, and AIR
 - b) Flowmeters: Electronic/Cascaded type for low and minimal flow anaesthesia
 - c) Oxygen Flush Facility: Conveniently placed flush switch (Flow: 25–70 L/min)
- 10) Vaporizers
 - a) Mounting capacity for two vaporizers, same make
 - b) Must be:
- 11) Temperature, pressure, and flow compensated
- 12) Isolated from gas flow in OFF position
- 13) Prevent simultaneous activation of both vaporizers
 - a) Included Vaporizers: 1 unit each of Sevoflurane and Isoflurane (latest generation, maintenance- and calibration-free)
 - b) Must comply with all safety norms and regulatory safety standards
- 14) Circuit System & Patient Interface
 - a) Operation Modes: Open & Closed circuit selection switch
 - b) Integrated circle absorber system:
- 15) Leak-proof, compact, autoclavable
- 16) Tool-free maintenance
- 17) Includes internal gas tubing, flow sensor wires, and other connections
- 18) Features Adjustable Pressure Limiting (APL) valve, airway pressure monitor, and Bag/Vent switch
 - a) Oxygen Sensor Provision:
- 19) Paramagnetic, Permanent, or Galvanic type
- 20) Sensor replacement free of cost during warranty/CMC period or up to 10 years, within 72 hours
 - a) CO₂ Absorber:
- 21) Single soda lime canister (≥800 grams)
- 22) Canister bypass feature during surgery without leakage
- 23) On-screen alert upon removal of canister
 - a) ACGO (Auxiliary Common Gas Outlet): Present
 - b) Work Area: Illuminated tabletop for arranging tools and drugs
 - c) Storage: Deep drawer(s)
 - d) Mobility: 4 castor wheels, front two with brakes

- 24) Power & Safety
 - a) Input Power: 100–240 VAC
 - b) Startup Pre-check Menu: Leakage and compliance testing
- 25) Emergency Bypass facility must be available
- 26) Built-In Ventilator (Integrated)
 - a) Capable of delivering precise ventilation for Adult, Pediatric, and Neonatal patients
 - b) Pneumatically driven & electronically controlled
- 27) Modes of Ventilation:
 - a) Volume Control (VCV)
 - b) Pressure Control (PCV)
 - c) SIMV + Pressure Support (PSV)
 - d) Pressure Support with Apnea Backup
 - e) Manual
- 28) Ventilation Parameter Ranges:
- 29) Parameter Range/Details
- 30) Tidal Volume (VCV) 20–1400 mL
- 31) Frequency 4–60 BPM or better
- 32) I:E Ratio 2:1 to 1:8
- 33) Inspiratory Pause 0–60%
- 34) Pressure Control 7–60 cm H₂O
- 35) Pressure Support 5–40 cm H₂O
- 36) Flow Trigger 0.5–10 LPM
- 37) PEEP Electronic, 4–20 cm H₂O or better
- 38) Sensor & Monitoring Features
 - a) Dual sensing capability at inhalation and exhalation ports
 - b) Sensors should be:
- 39) Reusable
- 40) Require no daily maintenance
 - a) Display of both Inspiratory and Expiratory tidal volume
 - b) Compensation:
- 41) Fresh gas flow compensation
- 42) Compliance compensation
- 43) Leak compensation (automatic)
- 44) Display & Measured Parameters
 - a) Minimum 10" or larger Touchscreen Colour TFT Display
- 45) Displayed Parameters:
 - a) Inspiratory Tidal Volume
 - b) Expiratory Tidal Volume
 - c) Minute Volume & MV-Spont
 - d) P_{peak}, PEEP, P_{plat}, P_{mean}, P_{min}
 - e) I:E Ratio
 - f) FiO₂
 - g) Compliance, Resistance
- 46) Waveforms & Loops :
 - a) At least 2 waveforms:
- 47) Pressure-Time

- 48) Volume-Time
 - a) At least 2 loops:
- 49) Pressure-Volume
- 50) Volume-Flow
- 51) Battery & Alarms
 - a) Internal Battery Backup: Minimum 60 minutes or more
 - b) Audio-Visual Alarms, categorized by priority
- 52) Alarm Settings For :
 - a) Tidal Volume
 - b) Minute Volume
 - c) Frequency
 - d) FiO₂
 - e) Airway Pressure
 - f) Apnea
 - g) Oxygen Failure
 - h) Mains Supply Failure
 - i) Low Battery
 - j) High Continuous Airway Pressure
- 53) Manufacturer/supplier should have ISO 13485 certificate for Quality standard.
- 54) IEC 60601-1 and EMI/EMC 61326-1 certification.
- 55) RoHS compliance is must.
- 56) Power input range 220-240V AC, 50Hz, fitted with Indian Plug.

9. Routine Flow Fluid Warmer

Item Code – ANES023

Item Description – Blood and Fluid Warmer, Patient warming system

Technical Specifications

- 1) Blood and Fluid Warmer
- 2) Should be able to warm fluid/blood to a temperature range of 37–40°C.
- 3) Should be able to maintain or warm fluid/blood at a flow rate of 2.5 L/min.
- 4) Should have a digital temperature display of fluid.
- 5) Should have inbuilt water tank or dry in-line heating system to warm the infused fluid/blood.
- 6) Should have a warm water column or heated sleeve up to the patient end to maintain temperature at the point of vein entry.
- 7) Alarms for disconnections, low water, and over-temperature.
- 8) At least 350 disposable tubing sets for fluid/blood to be supplied with each unit.
- 9) Should be suitable for both adult and pediatric patients.
- 10) Patient Warming System
- 11) Blankets for convective air patient warming system should be compatible with the basic warming system.
- 12) Blankets should be lightweight, puncture-resistant, and fluid-resistant.
- 13) Blankets should be latex-free, made of 2-ply material (non-woven outer layer and polyethylene inner layer).

- 14) Precision dye cut for even airflow and smooth surface.
- 15) Blankets should be disposable; 20 pediatric blankets to be supplied with each warming system.
- 16) Basic warming unit with disposable blankets to be included.
- 17) Fast warming capability: reach 38°C within 30 seconds.
- 18) Temperature settings: 30–34°C, 36–40°C, and 42–46°C.
- 19) Automatic step-down facility: after 45 minutes, temperature drops from high mode to medium mode.
- 20) HEPA filter with 0.2–0.3 micron filtration efficiency.
- 21) Multiple mounting options: cart, bedrail, IV pole, floor stand, and stainless steel trolley for easy ICU/Post-Op transport.
- 22) Auto power cut facility to control set temperature; sensors to prevent patient burns.
- 23) Hour meter to record total runtime.
- 24) Demonstration of the quoted equipment model is mandatory.
- 25) Consumables and spares must be available for the entire warranty and CMC period.
- 26) Rates of consumables and accessories to be quoted separately in the financial bid.

10. O.T Light LED with Camera

Item Code – OG023

Item Description – Double Combination OT Ceiling Light with centrally mounted camera

Technical Specifications

- 1) Should have the following features:
 - a) One Major and One Satellite Light Head of Different dimensions Reflector based LED Technology for better shadow resolution
 - b) Arrangement of LED in such a way that Shadow Free / Deep cavity illumination is achieved
 - c) Should have single color white LED's Special design to maximize the field of illumination and optimized illumination depth
 - d) Should have good laminar flow properties
 - e) Should have Aluminium Housing
 - f) ESG safety glass for simple and fast disinfection process 360 deg rotation of domes/lightheads / arms for unlimited positioning of light heads
 - g) MIS Lightning Feature on domes Multifunctional handle
 - h) Should have control panel on the cardanic arms
 - i) In case of LED failure, each individual LED module should be replaceable
 - j) Technical Data of Major Light Head:
 - i) Central illumination E_c (1mt) 160,000 lux
 - ii) Intensity Range (Lux) 30-100%
 - iii) Ambient Light for MIS: ≤ 5000 Lux
 - iv) Life time of light source (h) $\geq 60,000$
 - v) Light Field Diameter (mm) 180-300
 - vi) Depth of Field L1+L2 (mm) ≥ 850

- vii) CRI (Ra) Color range index ≥ 96 R9 (deep saturated red color index) ≥ 96
Colour Temp(K): 4500 K Fixed
- viii) Power consumption ≤ 70 W
- ix) Light Head Power Consumption 200 VA $\pm 10\%$
- x) Total Irradiance ≤ 650 W/m²
- xi) Temp increase at surgeons head < 1 deg C
- xii) LFI index(%): 16 LFI (measurement of particles) Class : 3.0 Power Supply-
Primary Voltage(V AC) 100-240 Rotation 360 deg Protection class according IEC 60601: 1 Suspension protection class: IP 30 Approx wt of each LH ≤ 18 kg
Light Diameter: ≥ 700 MM Hight/Depth of light head: ≤ 75 mm (without sterile handle) Technical Data of Satellite (1) Light Head Central illumination Ec (1mt) 140,000 lux each Intensity Range (Lux) 30-100% Ambient Light for MIS: ≤ 5000 Lux Life time of light source (h) $\geq 60,000$ Light Field Diameter (mm) 180-290 Depth of Field L1+L2 (mm) ≥ 900 CRI (Ra) Color range index ≥ 96 R9 (deep saturated red color index) ≥ 96 Colour Temp(K): 4500 K Fixed Power consumption ≤ 65 W Total Irradiance ≤ 550 W/m² Temp increase < 1 deg LFI (%) 14 LFI (measurement of particles) Class : 3.6 Power Supply-Primary Voltage(V AC) 100-240 Rotation 360 deg Classification according MDD: 1 Protection class according IEC 60601: 1 Suspension protection class: IP 30 Approx wt of each LH ≤ 15 kg Light Diameter: ≥ 600 MM Hight/Depth of light head: ≤ 75 mm Centrally Mounted Camera Magnification (Zoom) - 20x Optical zoom & 12x digital zoom Full HD 1920 x 1080 pixel
- k) It should have auto focus
- l) Manufactured by the same OEM as that of O.T. Light
- m) Attachment video output
 - i) Recording DVD recorder, Pen Drive
- n) Software Image & video recording software + CPU should be included
 - i) It should have facility for Still image & Video
 - ii) 24" LCD screen inside OT

11. Multiparameter Patient Monitor

Item Code – ANES026

Item Description – Multipara monitor with SpO₂, ECG, and NIBP

Technical Specifications

- 1) General Requirements
 - a) Equipment to be supplied with all standard accessories required to monitor all listed parameters.
 - b) Suitable for infants, children, and adolescents.
 - c) Wall-mountable, pivotable stand included.
 - d) Medical-grade TFT flat screen, slim size, with minimum 10-inch display.
 - e) Screen resolution: $\geq 1280 \times 1024$ pixels.
 - f) Color display with:
 - i) Large character size

- ii) Adjustable contrast and brightness
 - iii) User zoom function on any parameter
 - iv) Trace color customization per user preference
- 1) Waveform Display
- 2) Minimum 5 user-selectable channels.
- 3) Digital display of:
 - a) ECG
 - b) Heart Rate (HR)
 - c) Respiratory Rate (RR)
 - d) SpO₂
 - e) Non-Invasive Blood Pressure (NIBP)
- 4) ECG / Heart Rate Monitoring
 - a) At least 3-lead selectable ECG.
 - b) Built-in arrhythmia monitoring in all leads.
 - c) Inbuilt ST segment analysis and arrhythmia detection.
 - d) Heart rate range: 20–250 bpm.
 - e) Display sweep speeds: 12.5 and 25 mm/sec (user adjustable).
 - f) Averaging time: User selectable up to 8 seconds.
 - g) ECG amplitude: User adjustable.
 - h) Defibrillator protected.
- 5) Respiratory Rate Monitoring
 - a) Measured using transthoracic impedance via ECG leads.
 - b) Range: 6–150 breaths/min.
 - c) Accuracy: ± 2 bpm.
 - d) Sweep speeds: 6.25, 12.5, and 25 mm/sec (user adjustable).
 - e) Averaging time: User selectable up to 8 seconds.
 - f) Apnea alarm time: User selectable
- 6) Oxygen Saturation (SpO₂)
 - a) Range: 1–100%.
 - b) Accuracy: $\pm 2\%$ (within 40–100%).
 - c) Averaging time: User selectable up to 8 seconds.
 - d) Plethysmographic waveform display.
- 7) Non-Invasive Blood Pressure (NIBP)
 - a) Suitable for infants, children, and adolescents.
 - b) Oscillometric technique, microprocessor-controlled.
 - c) Measurement modes: Manual, Auto, and STAT.
 - d) Auto-deflate with overpressure protection.
 - e) Automatic zero reference after each reading.
 - f) Displays systolic, diastolic, and mean arterial pressure.
 - g) Unit: mmHg.
 - h) User-selectable time intervals.
- 8) Alarms – Audio & Visual
 - a) On-screen messages for:
 - b) High/Low Heart Rate.
 - c) High/Low Respiratory Rate.
 - d) Apnea (adjustable time: 5–20 sec).

- e) High/Low SpO₂.
 - f) High/Low Systolic, Diastolic, and Mean BP.
 - g) Probe failure.
 - h) Poor signal quality.
- 9) Power failure.
- 10) Trends and Data Storage
- a) Memory: ≥ 24 hours.
 - b) Data display interval: ≤ 20 seconds.
 - c) Display range: Last 30 minutes to 24 hours.
 - d) Display formats: Graphical and Tabular.
- 11) Power Requirements
- a) Power supply: 220–240 VAC, 50/60 Hz.
 - b) Internal rechargeable battery: Minimum 1-hour backup.
- 12) Essential Accessories (Per Monitor)
- a) ECG patient cables -2
 - b) SpO₂ extension cables -2
 - c) SpO₂ probes (Infant, Pediatric, Adult) -5 each
 - d) NIBP extension cables -2
 - e) Reusable NIBP cuffs (Infant, Pediatric, Adult) -2 each
- 13) Maintenance & Service Support
- a) Cost of consumables/accessories to be frozen during warranty and CMC periods.
 - b) Local service facility must be available.
 - c) Service provider must have manufacturer-recommended tools for preventive maintenance.
- 14) Documentation & Certification
- a) User, Technical, and Maintenance manuals in English.
 - b) Certificate of calibration and inspection.
 - c) List of essential spare parts and accessories with part numbers and prices.

12. Blood Auto Analyzer Machine

SPECIFICATIONS FOR BLOOD GAS ANALYSER

Bench top Blood Gas Analyser for use in critical care testing.

Meas

pH, pCO₂, pO₂ ,

Na⁺, K⁺, Ca⁺⁺, Cl⁻ , Haematocrit

Additional Calculated Parameters: up to 30 Parameters.

Maintenance-free electrodes.

Typically 70 microliters of sample for up to 10 parameters.

Sample throughput: 30 samples per hour.

Should accept whole blood and QC materials for blood gas measurements.

Should be able to generate Levy-Jennings plots from QC measurements.

Electrolytes should be measurable on whole blood, serum, plasma & Dialysate solutions.

Fully automatic liquid calibration for all parameters using internally generated gases. No gas cylinders. (Should be not cartridge base & gas cylinder base system).

Economy mode should be possible in automatic/manual modes.

In-built data management package to store patient results, QC results, calibration reports and self-diagnostics for up to two months.

HIS connectivity.

Built-in graphic thermal printer. Colour touch screen.

Bar code scanner for lab automation.

Operating conditions:

Instrument: +15 deg C to +33 deg C

Electrodes: +37 deg C

Solutions: +15 deg C to +33 deg C

Electrical Data:

Mains voltage: 220 volts

Frequency: 50 Hz

- Service center/Engineer should be in Chhattisgarh.
- FDA, CE, ISO approved
- Sample volume should be less than 100 microliters.

13. Endotracheal Tubes (Fr Gyn. Size-7, 7.5) (Plan Curved Duff)

(Fr Gyn. Size-7, 7.5) (Plan Curved Duff)

14. Stylet

Typically a malleable metal rod covered in a plastic sheath, it has a soft, rounded tip to minimize trauma.

15. Laproscopy Myoma Screen

Feature	Specification
Material	Surgical-grade stainless steel (some handles may be plastic)
Diameter	Typically 5 mm or 10 mm
Working Length	Commonly around 330 mm (ranging from 300 mm to 400 mm)
Usage	Reusable and fully autoclavable (sterilizable for safe re-use)
Handle Type	Often features an ergonomic or T-handle design for enhanced surgeon control
Certification	Often CE approved, FDA certified, and ISO certified

16. Bipolar

Feature	Common Specifications
Material	Stainless steel or titanium (titanium is lighter and non-glare)
Tip Size/Width	Varies widely: 0.2 mm, 0.4 mm, 0.5 mm, 0.7 mm, 1.0 mm, 2.0 mm, etc.
Total Length	Varies by application: 110 mm to 280 mm (e.g., 11.5 cm, 150 mm, 180 mm, 200 mm, 230 mm)
Forceps Shape	Straight, curved, or bayoneted (angled for better visibility in deep cavities)
Tip Style	Pointed, blunt, very fine, micro
Insulation	Fully insulated shaft, often with a non-stick coating (e.g., Teflon, noble metal alloy) on the tips to prevent tissue adherence and char buildup
Usability	Reusable (autoclavable) or single-use (sterile)

Connector Type Standard flat plug (EU type) or 2-pin plug (US type) to connect to an HF (high-frequency) generator cable
Compatibility Compatible with a wide range of industry-standard electrosurgical generators
Sterilization Reusable versions are typically autoclavable (max 134 °C)

17. Fowler's Bed

Specification

The Basic Fowler Bed has facility to raise the back upper leg sections. These sections are operated by crank mechanism provided at the end of the bed. The raising sections allow the nursing staff to change positions for various activities and clinical positions with relative ease. Overall approx dimension: 2095 mm L × 920 mm W × 600 mm H. Bed Frame approx dimension: 2095 mm L × 920 mm W. Pre-treated and powder coated finish. Supplied in SKD condition.

Optional Accessories for above bed

- Collapsible Type Safety Side Railing M. S. Powder Coated (Pair).
- Swing Down Type Rails (Pair)
- Collapsible Railing with Stainless Steel Tube (Pair)
- Four Section Mattress with 4" Thick PU Foam of 40 Density covered with PVC Rexine.
- Telescopic I.V. Pole
- Urine Bag Holder.

18. Eclampsia Bed

Specification for Eclampsia Bed

Overall Size: 2155mm L when extended x 1010 mm W x 520 mm To 880 mm H (Without Mattress). Three section high pressure laminated top. It shall have detachable laminated top for middle section and leg section for ease of cleaning.

Leg section can be telescoped under backrest for lithotomy position.

Backrest, trendelenburg & reverse trendelenburg and height adjustment position operated by Electro mechanical adjustment thru hand set; battery backup with inbuilt battery charger shall be provided. The hand set shall have indications for power on and the battery charge.

All electro mechanical actuators need to be compatible with class of IP 54.

The bed shall have work simultaneously on two actuators system for raising and lowering height. Bed frame made mainly from 60 x 30mm x 1.6mm mm thick ERW tube with proper support. This frame shall be fitted on the base frame mainly made of 60 x 30 x 1.6mm ERW tubes on various supporting links.

The base frame shall be mounted on 125mm dia non-rusting castor wheels with central and directional locking mechanism and pedal operated at the head end of the bed. Castor made from high grade non floor-staining synthetic materials with integrated thread guards. Wheel centre having precision ball bearing to run smoothly.

The bed shall have easily detachable moulded head & foot side panels. Four corner rubber buffers.

Bed shall have swing down type railings; Railing shall be made from non-rusting moulded material. These shall be fitted to the mattress support sections and should be able to raise and lock through spring lock mechanism through an operating lever.

S.S. saline rod with 12mm dia S.S. rod shall telescope in SS socket tube approx 15.8 mm dia x 18G welded on angular base bracket of 14G SS sheet. Nylon bracket provided to prevent colour

damage It could be placed at two different locations.

Patient's bearing down handgrips should be made of 22mm dia 18 G stainless steel tube fixed with PVC handle and should have location adjustability for patient's convenience.

A pair of upholstery aluminum lithotomic crutches mounted on S.S. rods.

Waste collecting removable stainless steel tray shall be provided at perennial recess.

It shall have foot support for high / low chair position.

The backrest and middle section shall have 75 mm thickness foam mattress P.U. 40 D. with rexine cover and zip. The leg section shall have 130mm thickness for propose alignment with other adjacent sections. The PU foam shall have high resilience and radiolucent property.

MS parts are passed through Shot blasting and 8 tank Pretreated & powder coated process

SS parts finished with Matt Polish.

19. Cryo Machine

Specification for Cryo Machine

- Control console with integrated 5ft.probe line yoke assembly
- Yoke
- Probes storage rack
- Choice of Cylinder Cart with 20 ft of scavenging hose (gas cylinder not included)

Features :

- Excellent thermal conductivity
- Temperature gauge for both cryo probe tip and thermocouple ensures adequate freeze with each treatment
- Rapid freeze and quick defrost
- Accurate and easy to read pressure gauge
- Probes with shaft connect at the handle for added patients safety
- Wide choice of probes for every lesion and location
- Console with integrated storage area for 5 tips
- Autoclavable probes and shaft
- System warranty 1 year.